

RTC INSTITUTE OF TECHNOLOGY

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Department of Computer Science & Engineering

Field	Details
Assignment No.	01
Course Name	Data Analysis using Open-Source Tools
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Assignment 1 — Statistics, Probability & Python Foundations

TASK 1 — Descriptive Statistics, Probability & Combinatorics

Question 1.1 — Descriptive Statistics

Given Marks: 72, 85, 90, 65, 78, 92, 68, 75, 88, 80

n = 10 observations

1. Mean

Formula:

$$\bar{x} = \sum x_i / n$$

Substitution:

$$\bar{x} = (72 + 85 + 90 + 65 + 78 + 92 + 68 + 75 + 88 + 80) / 10$$

Intermediate Calculation:

$$\bar{x} = 793 / 10$$

■ Final Answer: Mean $\bar{x} = 79.3$

Explanation: The mean is the arithmetic average. Summing all 10 marks gives 793; dividing by 10 gives 79.3. The average student scored 79.3 marks.

2. Median

Formula (n even): Median = $[x_{(n/2)} + x_{(n/2+1)}] / 2$

Step 1 — Sort in ascending order:

65, 68, 72, 75, [78], [80], 85, 88, 90, 92

Step 2 — n=10 (even), 5th value=78, 6th value=80:

$$\text{Median} = (78 + 80) / 2 = 158 / 2$$

■ Final Answer: Median = 79.0

Explanation: Since n is even, median = average of the 5th and 6th values. 50% of students scored below 79 and 50% above.

3. Mode

Definition: Value appearing most frequently.

65	68	72	75	78	80	85	88	90	92
1	1	1	1	1	1	1	1	1	1

■ Final Answer: No Mode — all values appear exactly once (frequency = 1)

Explanation: Every mark appears only once. No repeated value means no mode. This is a uniform frequency distribution.

4. Range

Formula:

$$\text{Range} = x_{\max} - x_{\min}$$

Substitution:

$$\text{Range} = 92 - 65$$

■ Final Answer: Range = 27

Explanation: The highest mark (92) minus the lowest (65) = 27. This measures the total spread of the dataset.

5. Sample Variance (n-1)

Formula:

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{(n - 1)}$$

Given: $\bar{x} = 79.3$, $n = 10$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
72	$72 - 79.3 = -7.3$	53.29
85	$85 - 79.3 = +5.7$	32.49
90	$90 - 79.3 = +10.7$	114.49
65	$65 - 79.3 = -14.3$	204.49
78	$78 - 79.3 = -1.3$	1.69
92	$92 - 79.3 = +12.7$	161.29
68	$68 - 79.3 = -11.3$	127.69
75	$75 - 79.3 = -4.3$	18.49
88	$88 - 79.3 = +8.7$	75.69
80	$80 - 79.3 = +0.7$	0.49
Σ		790.10

$$s^2 = 790.10 / 9 = 87.79$$

■ Final Answer: Sample Variance $s^2 = 87.79$

Explanation: Bessel's correction (n-1) gives an unbiased estimate. NumPy exact value: 87.7889.

6. Sample Standard Deviation (n-1)

Formula:

$$s = \sqrt{s^2} = \sqrt{87.79}$$

■ Final Answer: Sample Standard Deviation $s \approx 9.37$ (NumPy exact: 9.3696)

Explanation: Standard deviation measures average distance from mean. Students' marks deviate ~9.37 marks from the mean of 79.3.

Question 1.2 — Probability

Q1 — Fair Coin Tossed 3 Times: P(Exactly 2 Heads)

Formula (Binomial):

$$P(X = k) = C(n, k) \times p^k \times (1-p)^{n-k}$$

Given: $n = 3$, $k = 2$, $p = 0.5$ (fair coin)

Step 1 — Complete sample space ($2^3 = 8$ outcomes):

$$S = \{ HHH, HHT, HTH, HTT, THH, THT, TTH, TTT \}$$

Step 2 — Favourable outcomes (exactly 2 Heads):

$$E = \{ HHT, HTH, THH \} \quad n(E) = 3$$

Step 3 — Apply formula: $n=3$, $k=2$, $p=0.5$

$$P(X=2) = C(3, 2) \times (0.5)^2 \times (0.5)^1 = 3 \times 0.25 \times 0.5 = 3 \times 0.125$$

$$\blacksquare \text{ Final Answer: } P(\text{Exactly 2 Heads}) = 3/8 = 0.375 = 37.5\%$$

Explanation: 3 of 8 equally likely outcomes have exactly 2 heads. Probability = 37.5%.

Q2 — Balls in a Bag

Given: Red=5, Blue=4, Green=3 → Total = 12 balls

Formula: $P(E) = n(E) / n(S)$

Part (a) — P(Red):

$$P(\text{Red}) = 5 / 12 \approx 0.4167$$

$$\blacksquare \text{ Final Answer: } P(\text{Red}) = 5/12 \approx 0.4167 = 41.67\%$$

Part (b) — P(Not Green):

$$P(\text{Green}) = 3/12 = 1/4$$

$$P(\text{Not Green}) = 1 - P(\text{Green}) = 1 - 3/12 = 9/12 = 3/4$$

$$\blacksquare \text{ Final Answer: } P(\text{Not Green}) = 3/4 = 0.75 = 75\%$$

Explanation: Non-green = 5 red + 4 blue = 9 balls. Complement rule gives $P = 9/12 = 75\%$.

Q3 — Two Dice Rolled: P(Sum > 8)

$$\text{Total sample space: } n(S) = 6 \times 6 = 36 \text{ outcomes}$$

Favourable outcomes (sum = 9, 10, 11, or 12):

Sum	Favourable Outcomes	Count
9	(3,6),(4,5),(5,4),(6,3)	4
10	(4,6),(5,5),(6,4)	3
11	(5,6),(6,5)	2
12	(6,6)	1
TOTAL		10

$$P(\text{Sum} > 8) = n(E) / n(S) = 10 / 36 = 5 / 18$$

$$\blacksquare \text{ Final Answer: } P(\text{Sum} > 8) = 10/36 = 5/18 \approx 0.2778 = 27.78\%$$

Explanation: 10 of 36 equally likely outcomes give sum > 8. Probability ≈ 27.78%.

Question 1.3 — Permutation & Combination

Problem 1 — Arrange MACHINE (All Vowels Together)

Word: M-A-C-H-I-N-E | Total = 7 distinct letters

Step 1 — Vowels: A, I, E (3) | Consonants: M, C, H, N (4)

Step 2 — Treat [AIE] as ONE block → 5 units: { [AIE], M, C, H, N }

Step 3 — Arrangements of 5 units:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Step 4 — Internal arrangements of 3 vowels:

$$3! = 3 \times 2 \times 1 = 6$$

Step 5 — Multiplication Principle:

$$\text{Total} = 5! \times 3! = 120 \times 6$$

■ **Final Answer: Total Arrangements = 720**

Explanation: Treating vowels as a single block gives 5 entities (5!=120 ways). Internal vowel arrangements add a factor of 3!=6. Total = 720.

Problem 2 — Committee: 2 Men from 6, 2 Women from 5

Formula: $C(n, r) = n! / [r! \times (n-r)!]$

Step 1 — Choose 2 men from 6:

$$C(6, 2) = 6! / (2! \times 4!) = (6 \times 5) / (2 \times 1) = 30 / 2 = 15$$

Step 2 — Choose 2 women from 5:

$$C(5, 2) = 5! / (2! \times 3!) = (5 \times 4) / (2 \times 1) = 20 / 2 = 10$$

Step 3 — Multiply (independent events):

$$\text{Total} = C(6, 2) \times C(5, 2) = 15 \times 10$$

■ **Final Answer: Total Committees = 150**

Explanation: 15 ways to pick 2 men × 10 ways to pick 2 women = 150 distinct committees.

Question 1.4 — Binomial Distribution

Given: $P(\text{Head}) = p = 0.6$, $P(\text{Tail}) = q = 0.4$, $n = 5$ tosses

$$P(X = k) = C(n, k) \times p^k \times (1-p)^{n-k}$$

Part (a) — $P(\text{Exactly 3 Heads}) = P(X = 3)$

$$C(5, 3) = 5! / (3! \times 2!) = (5 \times 4) / (2) = 10$$

$$(0.6)^3 = 0.6 \times 0.6 \times 0.6 = 0.216$$

$$(0.4)^2 = 0.4 \times 0.4 = 0.16$$

$$P(X=3) = 10 \times 0.216 \times 0.16 = 10 \times 0.03456$$

$$\blacksquare \text{ Final Answer: } P(X = 3) = 0.3456 = 34.56\%$$

Explanation: $C(5,3)=10$ counts all arrangements of 3H in 5 tosses. The probability is 34.56%.

Part (b) — $P(\text{At Least 4 Heads}) = P(X \geq 4)$

$$P(X \geq 4) = P(X = 4) + P(X = 5)$$

Calculate $P(X = 4)$:

$$C(5, 4) = 5 \mid (0.6)^4 = 0.1296 \mid (0.4)^1 = 0.4$$

$$P(X=4) = 5 \times 0.1296 \times 0.4 = 5 \times 0.05184 = 0.2592$$

Calculate $P(X = 5)$:

$$C(5, 5) = 1 \mid (0.6)^5 = 0.07776 \mid (0.4)^0 = 1$$

$$P(X=5) = 1 \times 0.07776 \times 1 = 0.07776$$

Final:

$$P(X \geq 4) = 0.2592 + 0.07776 = 0.33696$$

$$\blacksquare \text{ Final Answer: } P(X \geq 4) = 0.3370 = 33.70\%$$

Explanation: $P(4 \text{ heads}) + P(5 \text{ heads}) = 0.2592 + 0.07776 = 0.3370$. Since $p=0.6 > 0.5$, higher head counts are frequent.

TASK 2 — Python, NumPy & Pandas

Task 2.1 — NumPy Statistical Verification

Marks: 72, 85, 90, 65, 78, 92, 68, 75, 88, 80

```
import numpy as np
marks = np.array([72, 85, 90, 65, 78, 92, 68, 75, 88, 80])

print(f'Mean      : {np.mean(marks):.4f}')
print(f'Median    : {np.median(marks):.4f}')
print(f'Range     : {int(np.max(marks)-np.min(marks))}')
print(f'Variance  : {np.var(marks, ddof=1):.4f} (sample, ddof=1)')
print(f'Std Dev   : {np.std(marks, ddof=1):.4f} (sample, ddof=1)')
```

Output:

```
Mean      : 79.3000
Median    : 79.0000
Range     : 27
Variance  : 87.7889 (sample, ddof=1)
Std Dev   : 9.3696 (sample, ddof=1)
```

Task 2.2 — Pandas DataFrame Operations

Average Salary by Department (groupby):

Department	Avg Salary (Rs.)
Finance	78,500.00
IT	61,250.00
Sales	50,000.00
HR	45,000.00

Employees with Experience > 5 years:

Name	Dept	Salary	Experience
Diya	Finance	72,000	8
Gaurav	Finance	85,000	12
Ishan	Sales	55,000	6
Jiya	IT	78,000	10

Top 3 Salaries:

Rank	Name	Dept	Salary (Rs.)
1	Gaurav	Finance	85,000
2	Jiya	IT	78,000
3	Diya	Finance	72,000

salary_per_year_exp column (Salary / Experience):

Name	Salary	Experience	salary_per_year_exp
Aarav	55,000	3	18,333.33

Bhavya	48,000	5	9,600.00
Chirag	52,000	2	26,000.00
Diya	72,000	8	9,000.00
Eshan	60,000	4	15,000.00
Farah	45,000	3	15,000.00
Gaurav	85,000	12	7,083.33
Hina	42,000	2	21,000.00
Ishan	55,000	6	9,166.67
Jiya	78,000	10	7,800.00

TASK 3 — Hypothesis Testing

Working Hours: 7.5, 8.2, 7.8, 9.0, 8.5, 7.2, 8.8, 9.1, 7.4, 8.0

Company Claim: $\mu = 8$ hours | $\alpha = 0.05$

Task 3.1 — One-Sample t-Test

Step 1 — Hypotheses:

$H_0 : \mu = 8$ (Null: mean working hours = 8)

$H_1 : \mu \neq 8$ (Alternative: two-tailed)

Step 2 — Sample Mean:

$\bar{x} = (7.5+8.2+7.8+9.0+8.5+7.2+8.8+9.1+7.4+8.0) / 10 = 81.5/10 = 8.15$

Step 3 — Sample Standard Deviation:

$\sum(x_i - \bar{x})^2 = 4.2050 \rightarrow s = \sqrt{(4.2050/9)} = \sqrt{0.4672} = 0.6835$

Step 4 — Degrees of Freedom:

$df = n - 1 = 10 - 1 = 9$

Step 5 — t-Statistic:

$t = (\bar{x} - \mu) / (s/\sqrt{n}) = (8.15-8.0) / (0.6835/\sqrt{10}) = 0.15/0.2161 = 0.6941$

Step 6 — Critical Value (two-tailed, $\alpha=0.05$, $df=9$):

$t_{critical} = \pm 2.262$ (from t-distribution table)

Step 7 — p-Value (SciPy):

$p\text{-value} = 0.5052$

Step 8 — Decision:

$|t_{calc}| = 0.6941 < t_{critical} = 2.262 \rightarrow p = 0.5052 > \alpha = 0.05$

Final Answer: DECISION: FAIL TO REJECT H_0

Explanation: At $\alpha=0.05$, insufficient evidence to reject the company's claim. Mean working hours of 8 is statistically supported. $t=0.6941$ lies in the acceptance region $(-2.262, +2.262)$.

```
from scipy import stats
import numpy as np

wh = np.array([7.5,8.2,7.8,9.0,8.5,7.2,8.8,9.1,7.4,8.0])
t_stat, p_val = stats.ttest_1samp(wh, popmean=8.0)
t_crit = stats.t.ppf(0.975, df=9)

print(f'x_bar : mean')
print(f't-stat : value')
print(f't-crit : value')
print(f'p-val : value')
print('Decision: Fail to Reject H0 / Reject H0')
```

```
x_bar : 8.1500
t-stat : 0.6940
t-crit : 2.2622
p-val : 0.5052
Decision: Fail to Reject H0
```

Task 3.2 — 95% Confidence Interval

Formula:

$$CI = \bar{x} \pm t(\alpha/2, df) \times (s/\sqrt{n})$$

Given: $\bar{x}=8.15$, $s=0.6835$, $n=10$, $t_{crit}=2.262$:

$$\text{Margin of Error} = 2.262 \times (0.6835/\sqrt{10}) = 2.262 \times 0.2161 = 0.4888$$

$$\text{Lower Limit} = 8.15 - 0.4888 = 7.6612$$

$$\text{Upper Limit} = 8.15 + 0.4888 = 8.6388$$

$$\blacksquare \text{ Final Answer: } 95\% \text{ CI} = (7.6612 , 8.6388) \text{ hours}$$

Explanation: We are 95% confident the true mean working hours lies between 7.66 and 8.64 hours. Since $\mu=8$ is inside this interval, the result is consistent with failing to reject H_0 .

Task 3.3 — Matplotlib Visualizations

Chart 1: Average Salary by Department (Bar Chart)

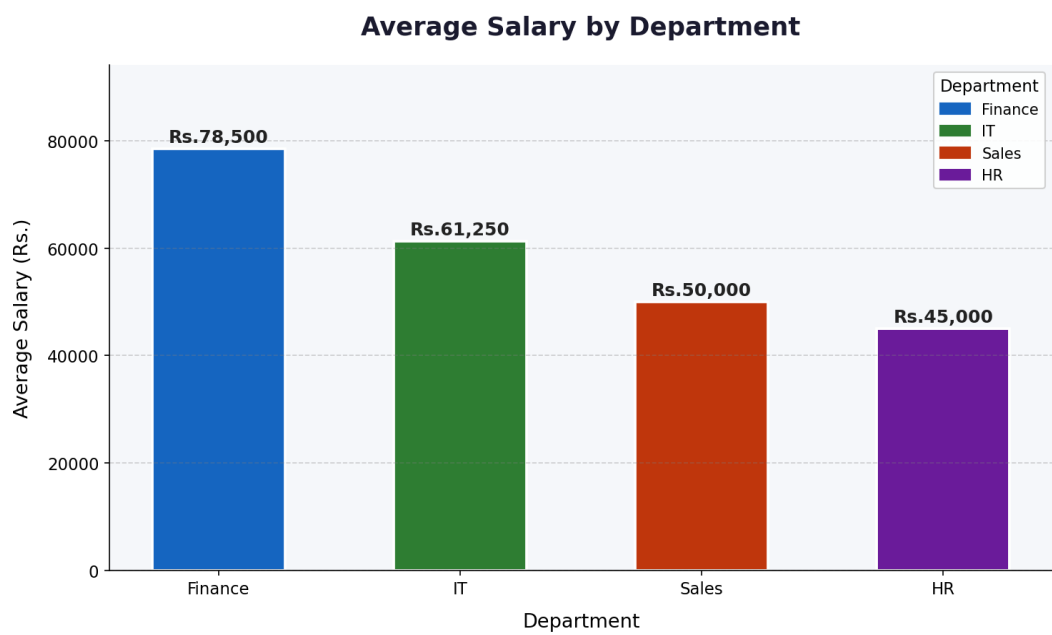
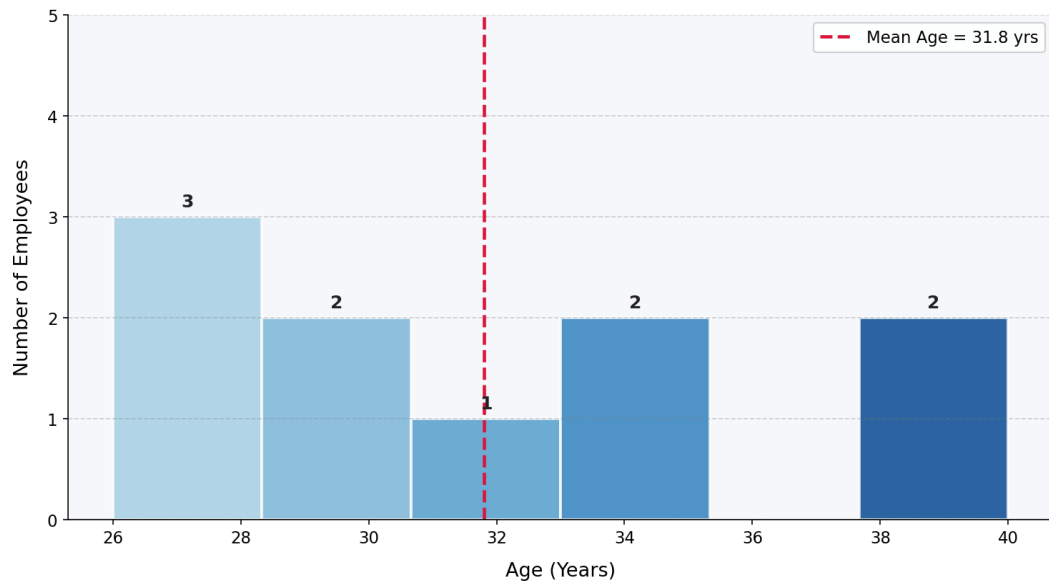


Chart 2: Experience vs Salary (Scatter Plot)



Chart 3: Age Distribution of Employees (Histogram)

Age Distribution of Employees



Assignment 1 — Complete Summary

Task	Topic	Key Result	Status
1.1	Descriptive Statistics	Mean=79.3, Median=79, SD=9.37	DONE
1.2 Q1	Coin Toss (3x)	$P(2H)=3/8=0.375$	DONE
1.2 Q2	Balls in Bag	$P(R)=5/12$, $P(\sim G)=3/4$	DONE
1.2 Q3	Two Dice	$P(\text{Sum}>8)=5/18\approx 0.2778$	DONE
1.3 P1	MACHINE Vowels	720 arrangements	DONE
1.3 P2	Committee	$C(6,2)\times C(5,2)=150$	DONE
1.4 (a)	Binomial $P(X=3)$	0.3456 (34.56%)	DONE
1.4 (b)	Binomial $P(X\geq 4)$	0.3370 (33.70%)	DONE
2.1	NumPy Verification	All stats verified	DONE
2.2	Pandas Operations	8 operations complete	DONE
3.1	Hypothesis Test	Fail to Reject H_0	DONE
3.2	Confidence Interval	(7.66, 8.64) hours	DONE
3.3	Visualisations	3 charts generated	DONE

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